Home Appliances Protector

Objective:

I built an ESP32-based voltage protector device that will save home appliances. Especially when the power grid voltage surges or over-current faults occur (like voltage goes too high or too low, and current goes too high)

Design Approach:

- The design was done in two phases: in the high-voltage part and the low-voltage part. For simulation I used Wokwi.com. Because of some electrical part limitations in Wokwi. I used some electronics parts instead. (board-esp32-devkit-c-v4)

Component Description:

Real Product	**Used in simulation**	**In ESP -32**	**Working Principle**
ZMPT101B, voltage sensor, ACS712-20A current sensor	Potentiometer	GPIO34	It will get voltage from grid and work as transformer convert(220v to 3.3v)
Contactor + appliance	Relay module + yellow LED	GPIO 35	It will work as a switch
Buzzer, LEDs	Buzzer, red LED, green LED	GPIO, 35,25,32	It works as an indicator.
User interface	LCD 16x2 I2C, menu/up/down buttons	GPIO 21,22, 14,13,4	LCD for changes display, buttons for change settings

Logic:

- If v > 250, v < 180, or current > 10, then the relay opens immediately. Load will be disconnected
- Load comes back when voltage stays in a 5-volt safe margin (185 V - 245 V) for reconnected delay about 5 s

Changing Settings:

- Three Buttons (Menu, Up, Down); In Menu (Set Over-Volt, Set Under-Volt, Set Max Current), Up (+5 V for voltage, +0.5 A for current), Down (-5 V for voltage, -0.5 V for current)

Result:

- Voltage Over: Above 250 V—Fault: Over voltage, Load state: Disconnected
- Voltage under 180 V—Fault: under voltage, Load state: Disconnected
- Voltage between 180 and 250 Fault: Normal, Load State: Connected (after 5 seconds stay this voltage range)
- Here Load can be used for fans, motors, etc., but in Wokwi simulation used LED (blue)

Limitation :

- One channel output (for example it is for one relay channel like fan)

Future Plan:

- Multiple channel output (multiple relay channel like fan,motor, AC etc)